Nuclear Radiation

During the Program

Discovery

1. What is radioactivity?

_____

_____

_____

2. Ernest Rutherford discovered that there were three different types of radiations. What are they?

_____

_____

_____

3. What are some of the uses of radioactive materials today?

_____

_____

_____

4. What are isotopes?

_____

_____

_____

5. a) What are the three main isotopes of carbon?

_____

b) What is the difference between them?

_____

_____

_____

Nuclear stability

6. What provides the 'strong nuclear force' in an atom's nucleus?

_____

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7. Complete this sentence:

   Atoms are radioactive if there is an imbalance between the number of:

   _____

8. a) What is a radioisotope?

   _____

   b) How is carbon-14 used to determine the age of samples?

   _____

   _____

   c) Give an example of:

   - a natural radioisotope and its use

   _____

   _____

   - an artificially produced radioisotope and its use

   _____

   _____

9. a) What is a positron and what is its symbol?

   _____

   b) Fill in the missing words to complete this sentence:

   In positron emission tomography (PET), positrons and electrons annihilate each other, producing penetrating _____ rays which can be used in _____diagnosis to produce _____ images.

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Types of radiation

10. Complete the following table summarising the properties of the various types of radiation:

Types of Radiation	alpha	beta	gamma
Description			
Mass (heavy, light, none)			
Charge (none, weak, strong)			
Ionising ability (strong, weak, none)			
Effect of electric & magnetic fields (strong, weak, none)			
Penetration (describe)			
Real life example			

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11. Write decay equations for the following:

a) alpha decay of americium-241, $^{241}_{95}\mathrm{Am}$

_____

b) beta decay of cobalt-60, $^{60}_{27}\mathrm{Co}$ (1 line)

_____

c) alpha and gamma decay of uranium-238, $^{238}_{92}U$

_____

Detection, dosage and effects of radiation

12. Why are film badges, dosimeters and thermo-luminescent dosimeters (TLDs) important?

_____

_____

_____

13. What effects can radiation have on human cells?

_____

_____

_____

14. Complete the following radiation dose and dose equivalent calculations.

Mass of Person (kg)	Energy (J)	Radiation Dose (Jkg^{-1}, or Gy)	Type of Radiation	Dose Equivalent (Sv)
50	100		gamma	
50	100		alpha	

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15. Explain the difference between the terms radiation dose (Gy) and dose equivalent (Sv).

_____

_____

_____

Half life

16. What do we mean by the term ‘half life’?

_____

_____

_____

17. What is a decay series and how is it produced?

_____

_____

_____

18. What are some of the positive and negative effects of radiation?

_____

_____

_____

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